# Design & Analysis of Algorithms

Lab 1: Algorithm Performance Analysis Workbench

Name: Prachi  Manwal
Roll No: 2501730365
Email: 2501730365@krmu.edu.in

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### Q1. 1.4 Algorithm Execution Observation Table  (5 marks)

```
import math


def generate_execution_observation_table(sizes):
    # Precompute recursive Fibonacci call counts: C(n) = 1 + C(n-1) + C(n-2)
    fib_calls = [0] * 35
    fib_calls[0] = 1
    fib_calls[1] = 1
    for i in range(2, 35):
        fib_calls[i] = 1 + fib_calls[i - 1] + fib_calls[i - 2]

    # Initialize the output list with the required header lines
    table = [
        "Algorithm Execution Observation Table",
        "InputSize RecursiveFactorial IterativeFactorial RecursiveFibonacci "
        "IterativeFibonacci LinearSearch BinarySearch BubbleSort InsertionSort",
    ]

    # Calculate deterministic counts for each input size
    for n in sizes:
        rec_fact = n + 1
        iter_fact = n
        rec_fib = fib_calls[n]
        iter_fib = n
        linear_search = n
        binary_search = math.floor(math.log2(n)) + 1
        bubble_sort = n * (n - 1) // 2
        insertion_sort = n * (n - 1) // 2

        row = (
            f"{n} {rec_fact} {iter_fact} {rec_fib} {iter_fib} "
            f"{linear_search} {binary_search} {bubble_sort} {insertion_sort}"
        )
        table.append(row)

    return table
```